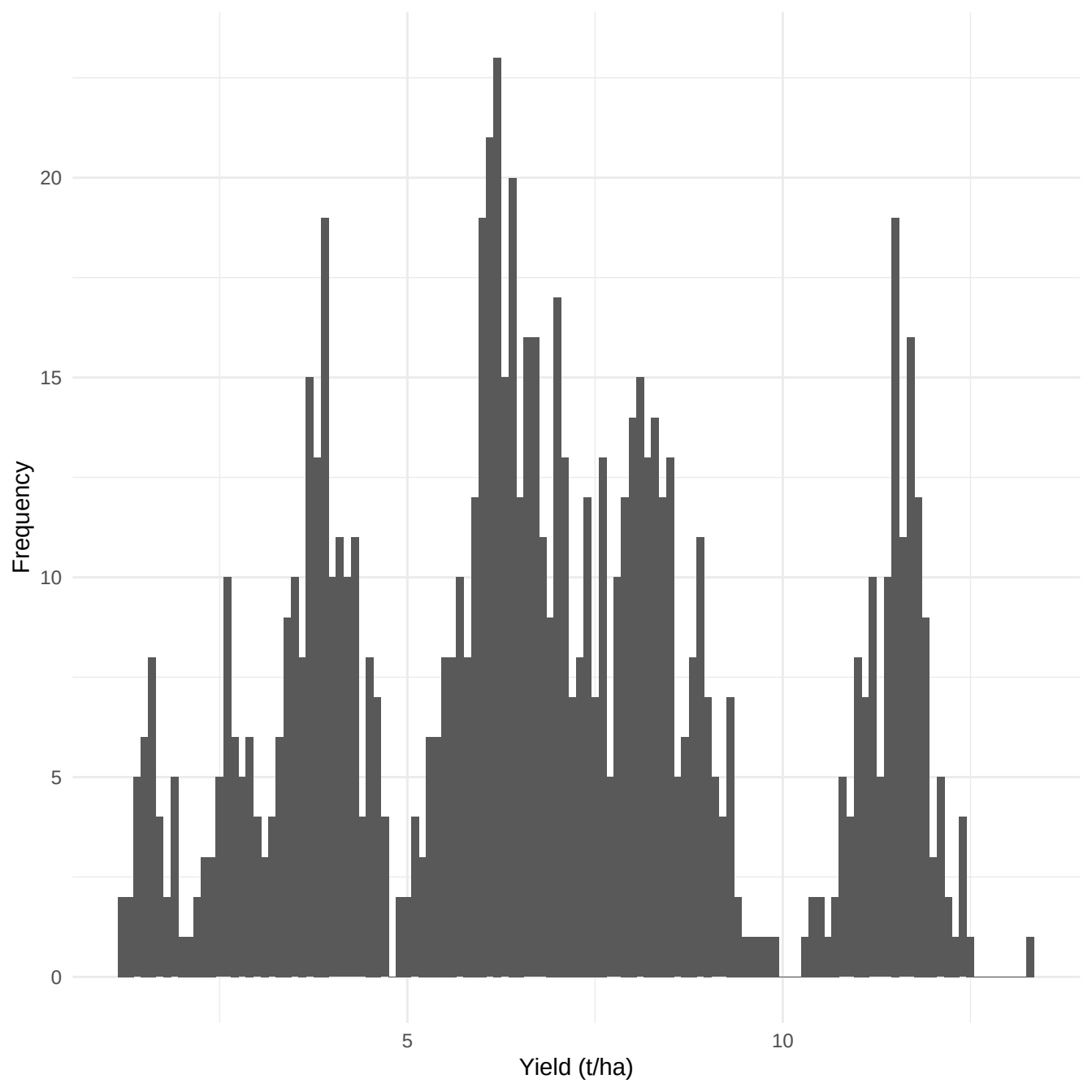
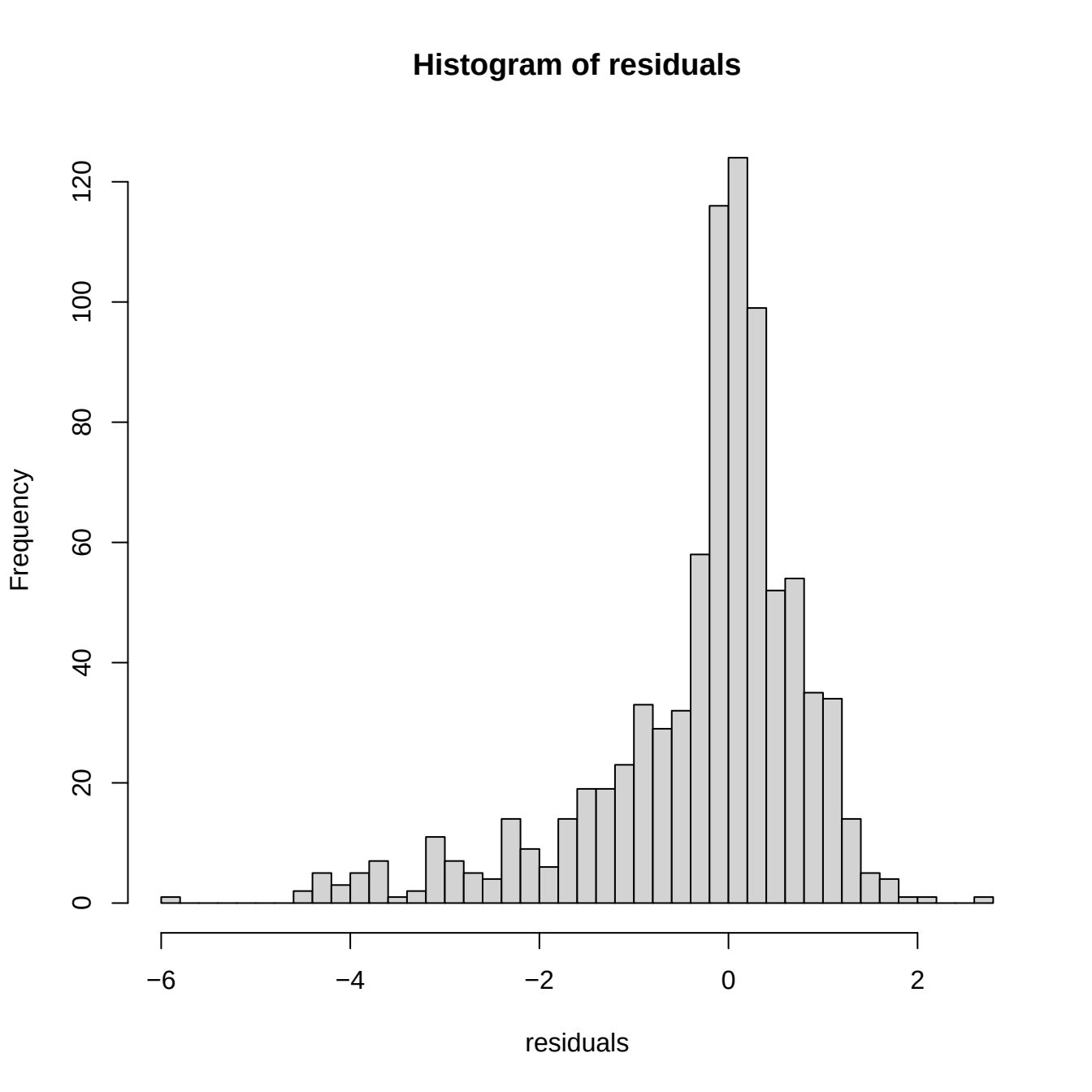
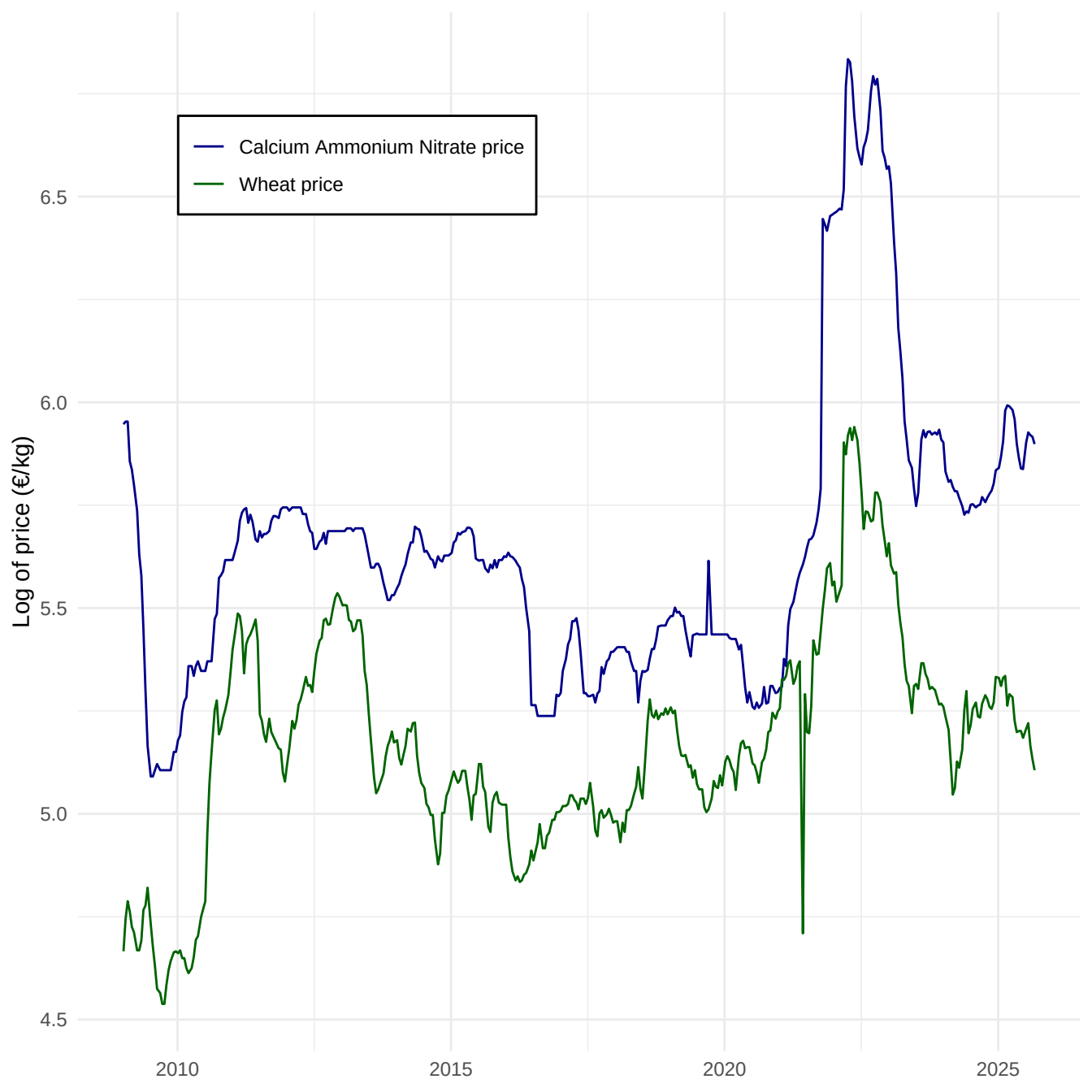
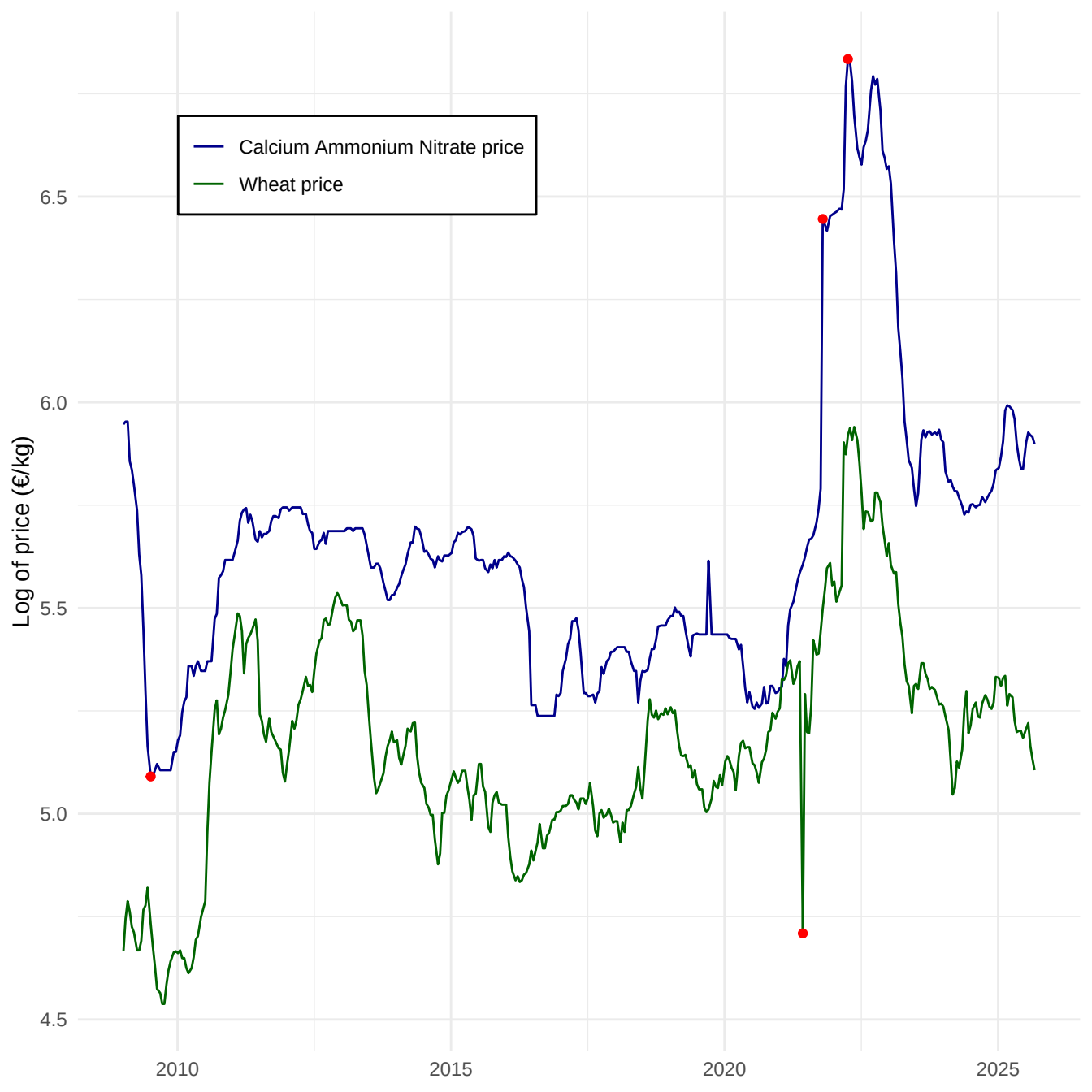
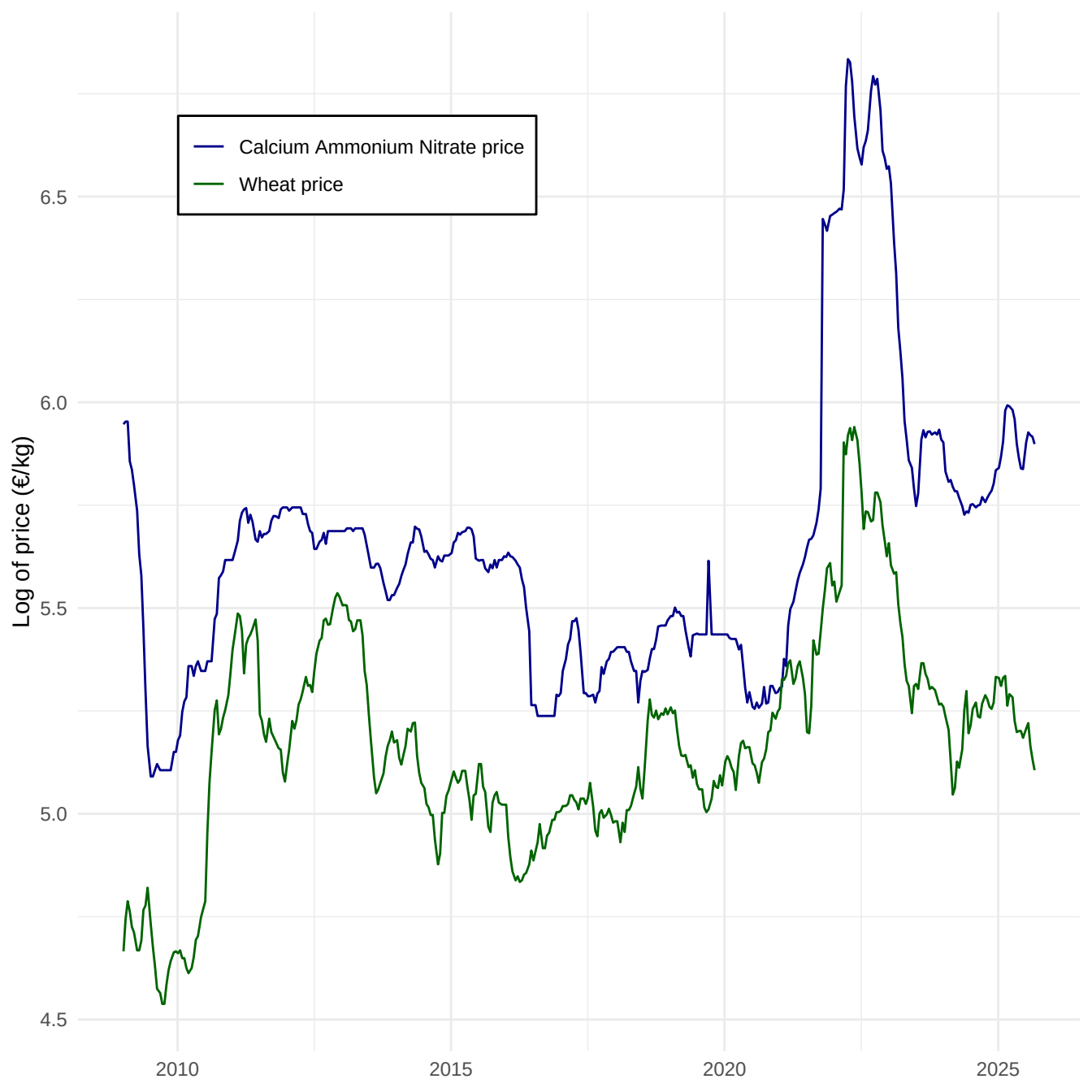
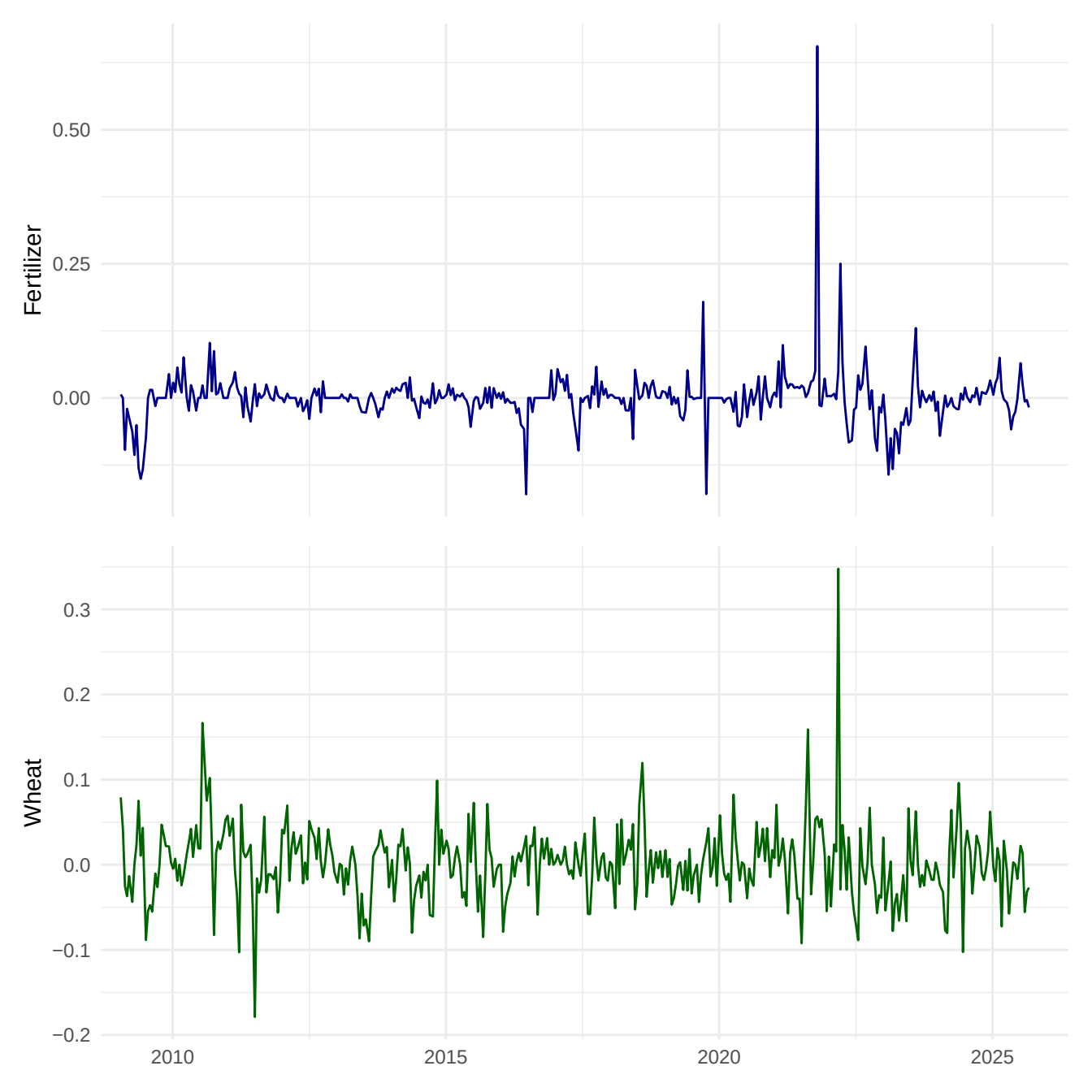

Frequency
20
15
10
5
0
5
10
Yield (t/ha)

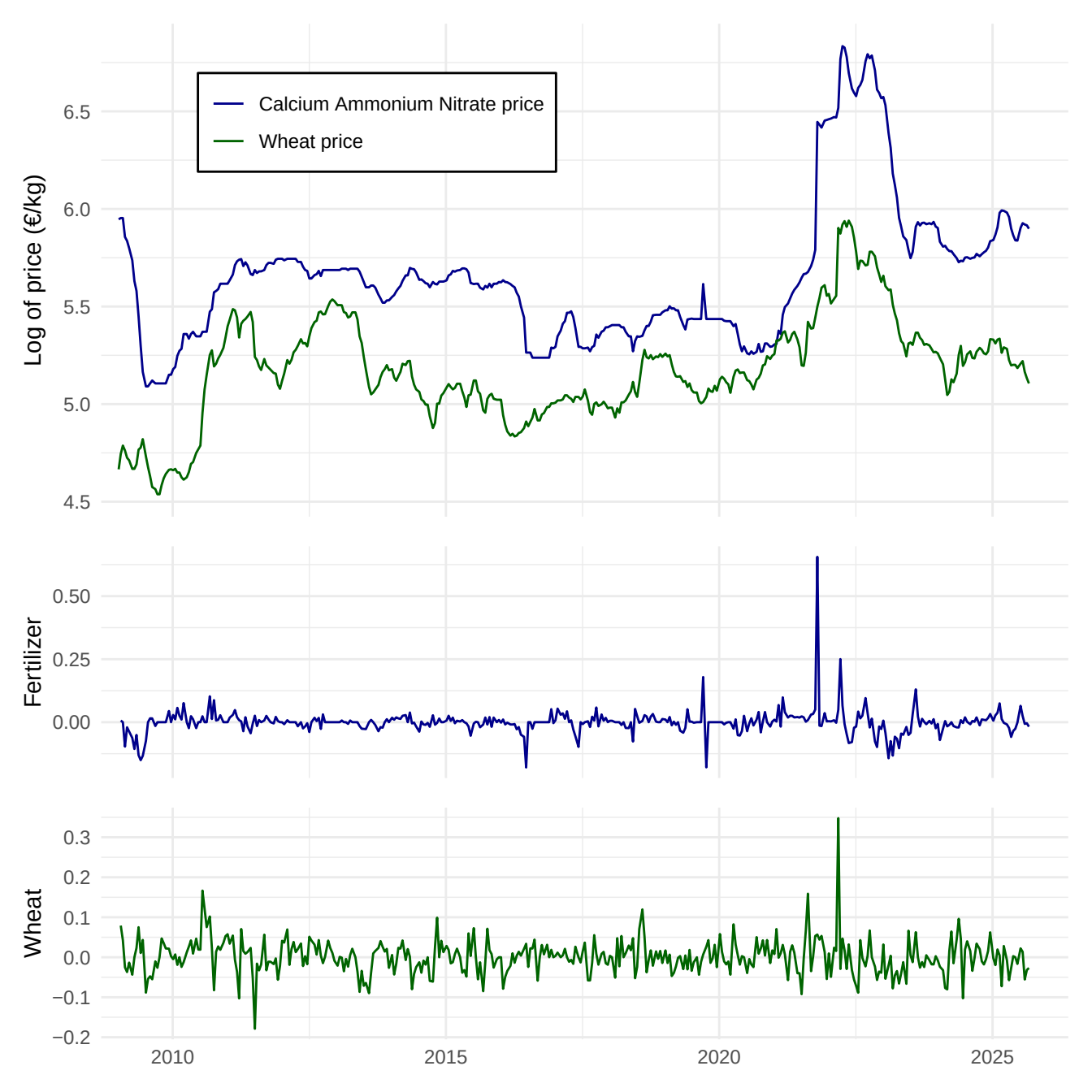

Histogram of residuals
Frequency
0
20
40
60
80
100
120
−6
−4
−2
0
2
residuals

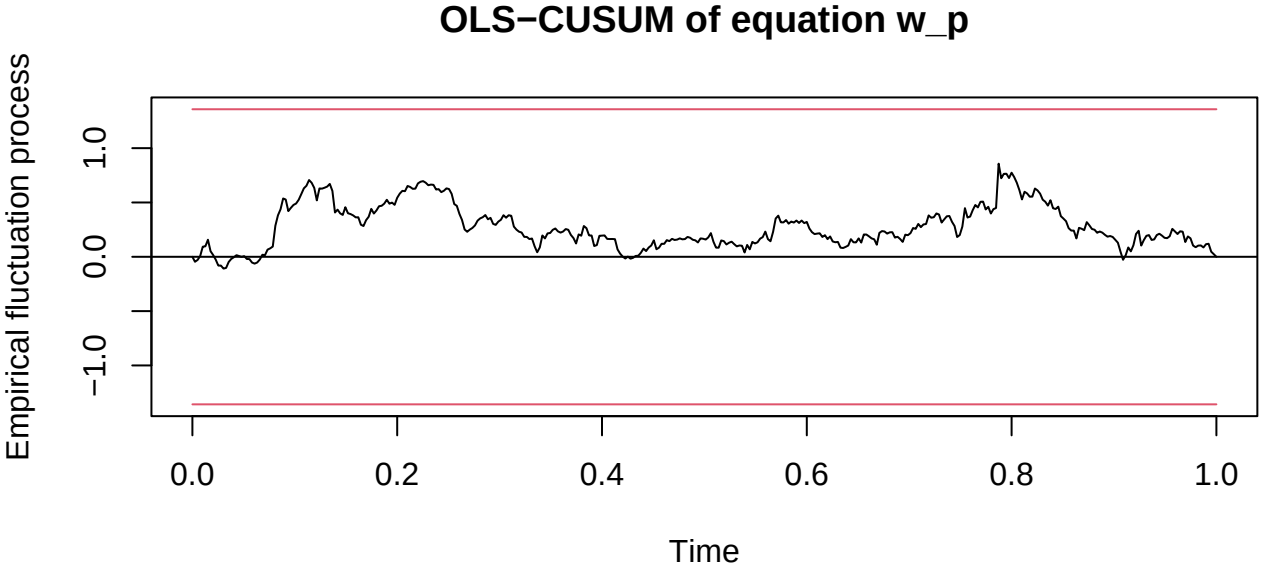

Calcium Ammonium Nitrate price
Wheat price
Log of price (€/kg)
6.5
6.0
5.5
5.0
4.5
2010
2015
2020
2025

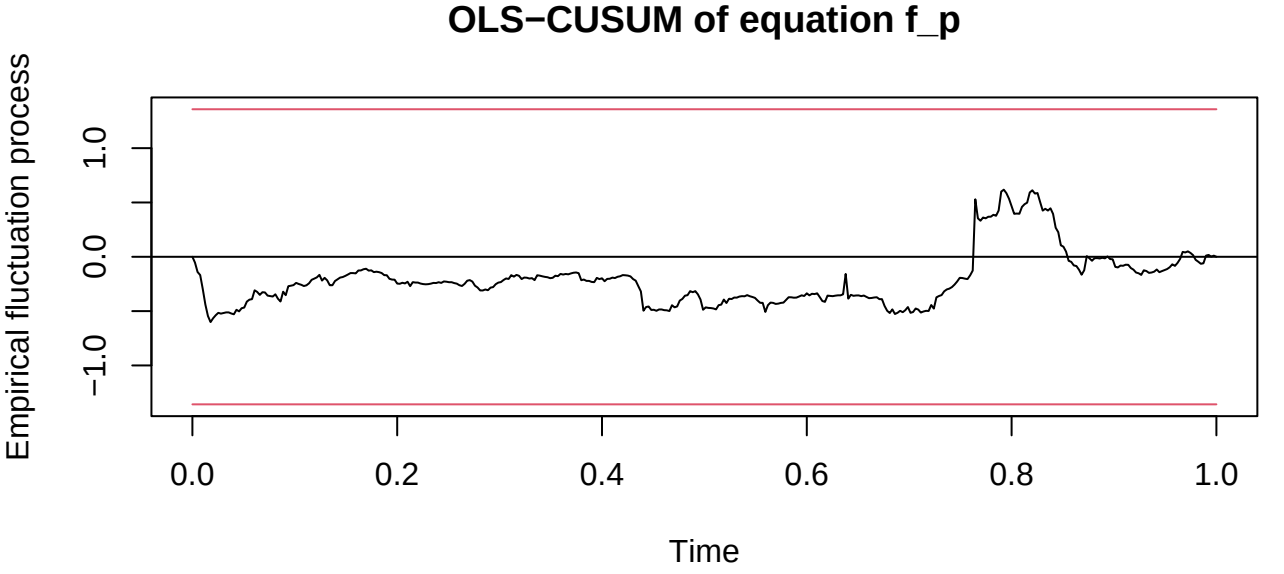

Calcium Ammonium Nitrate price
Wheat price
Log of price (€/kg)
6.5
6.0
5.5
5.0
4.5
2010
2015
2020
2025

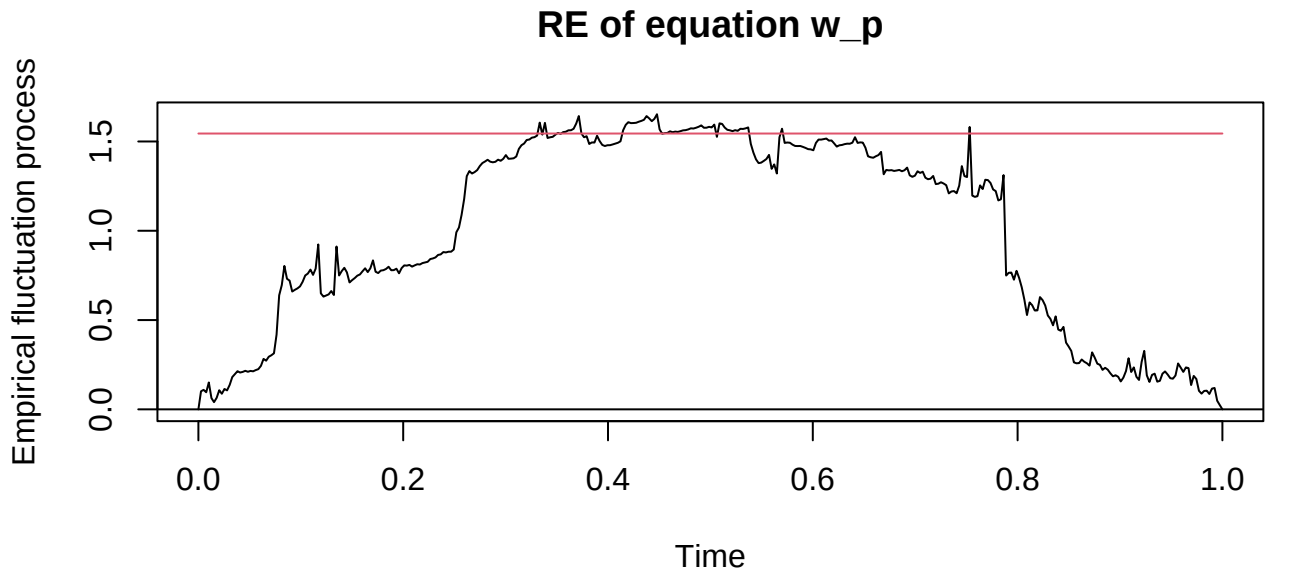

Calcium Ammonium Nitrate price
Wheat price
Log of price (€/kg)
6.5
6.0
5.5
5.0
4.5
2010
2015
2020
2025

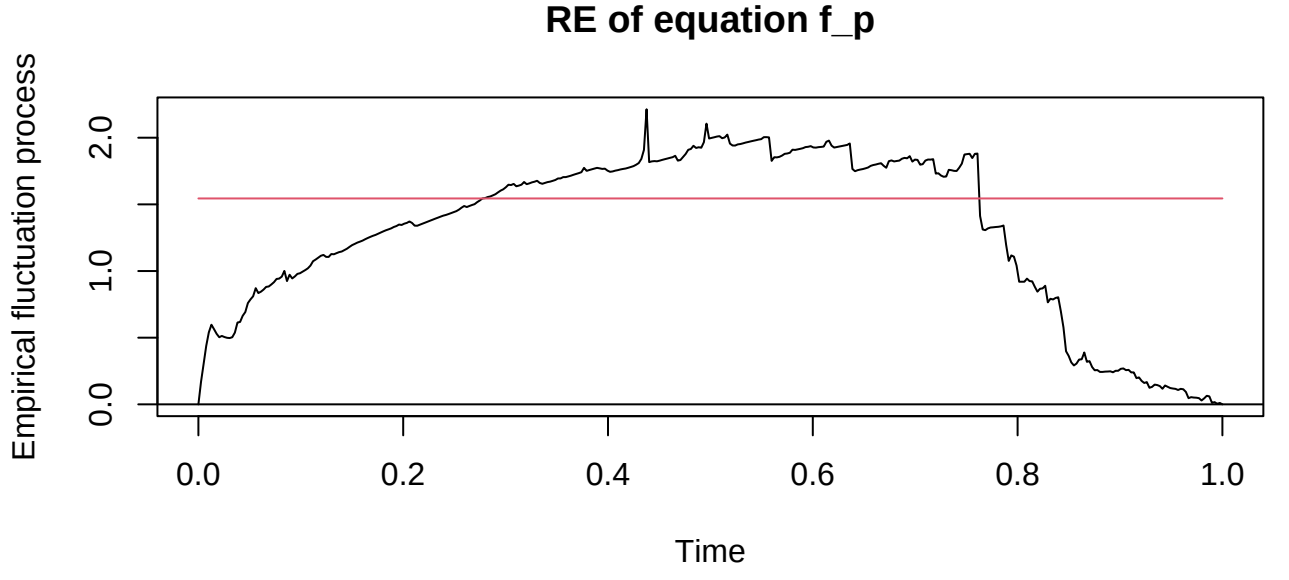

Fertilizer
0.50
0.25
0.00
Wheat
0.3
0.2
0.1
0.0
−0.1
−0.2
2010
2015
2020
2025

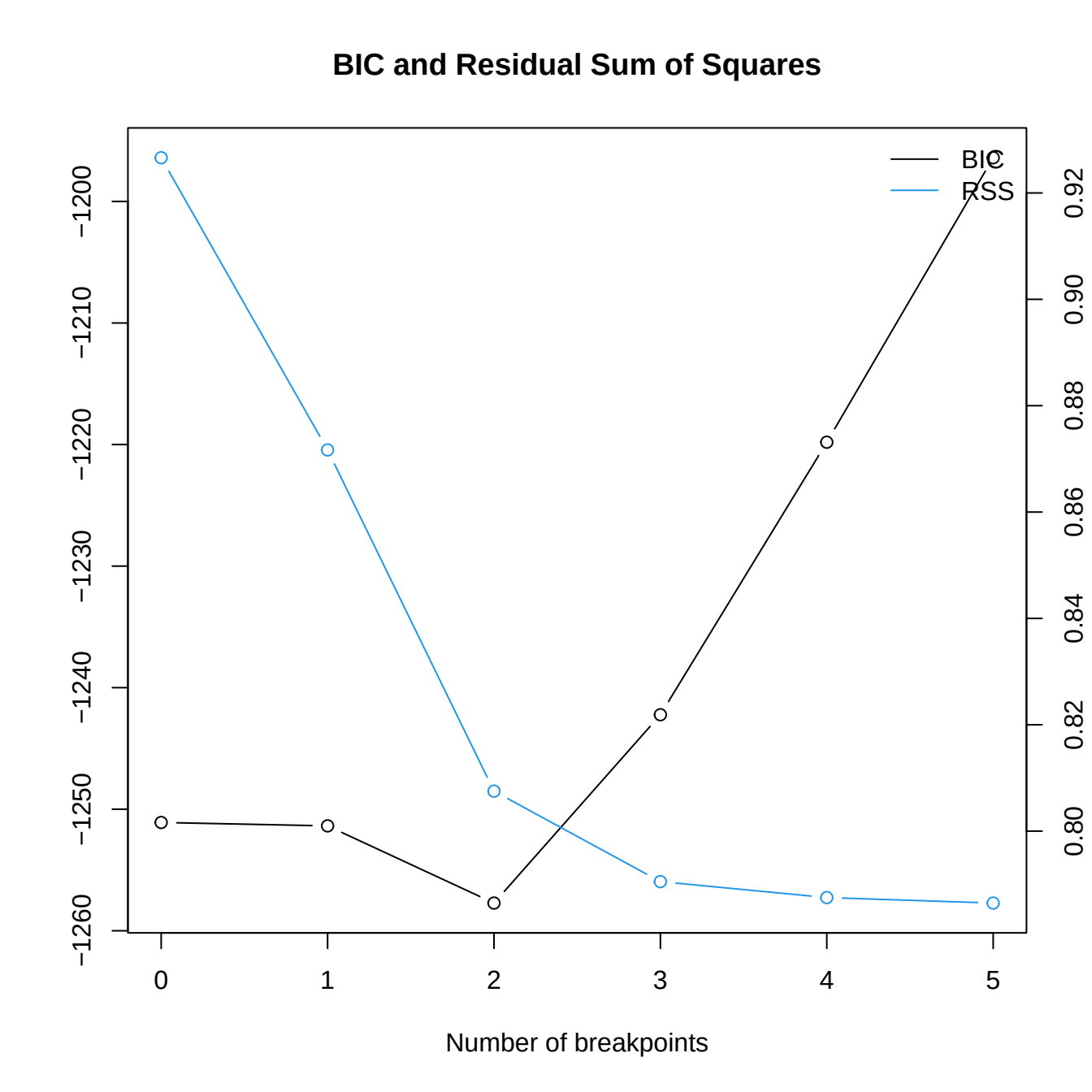

Calcium Ammonium Nitrate price
Wheat price
Log of price (€/kg)
6.5
6.0
5.5
5.0
4.5
Fertilizer
0.50
0.25
0.00
Wheat
0.3
0.2
0.1
0.0
−0.1
−0.2
2010
2015
2020
2025

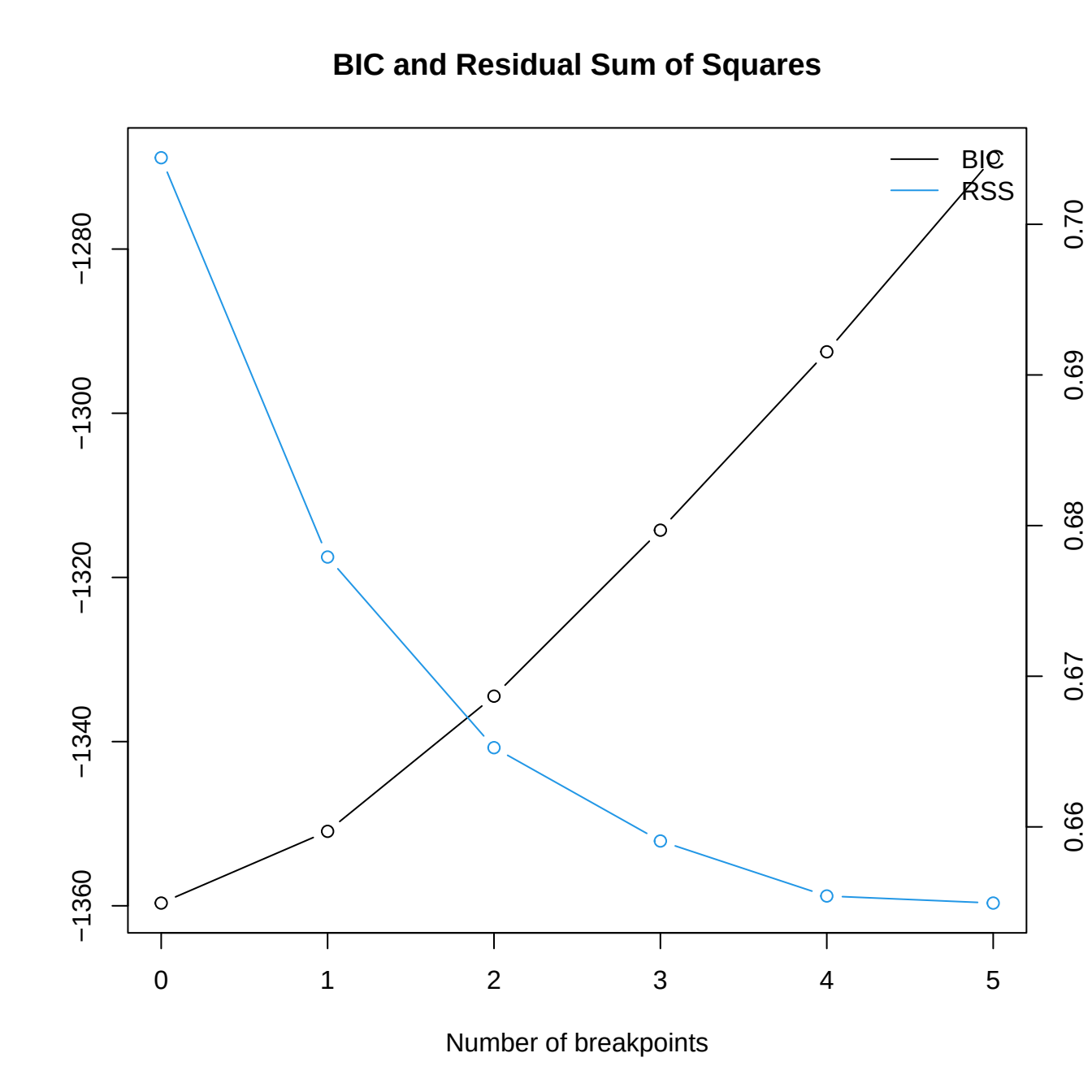

OLS−CUSUM of equation w_p
Empirical fluctuation process
1.0
0.0
−1.0
0.0
0.2
0.4
0.6
0.8
1.0
Time

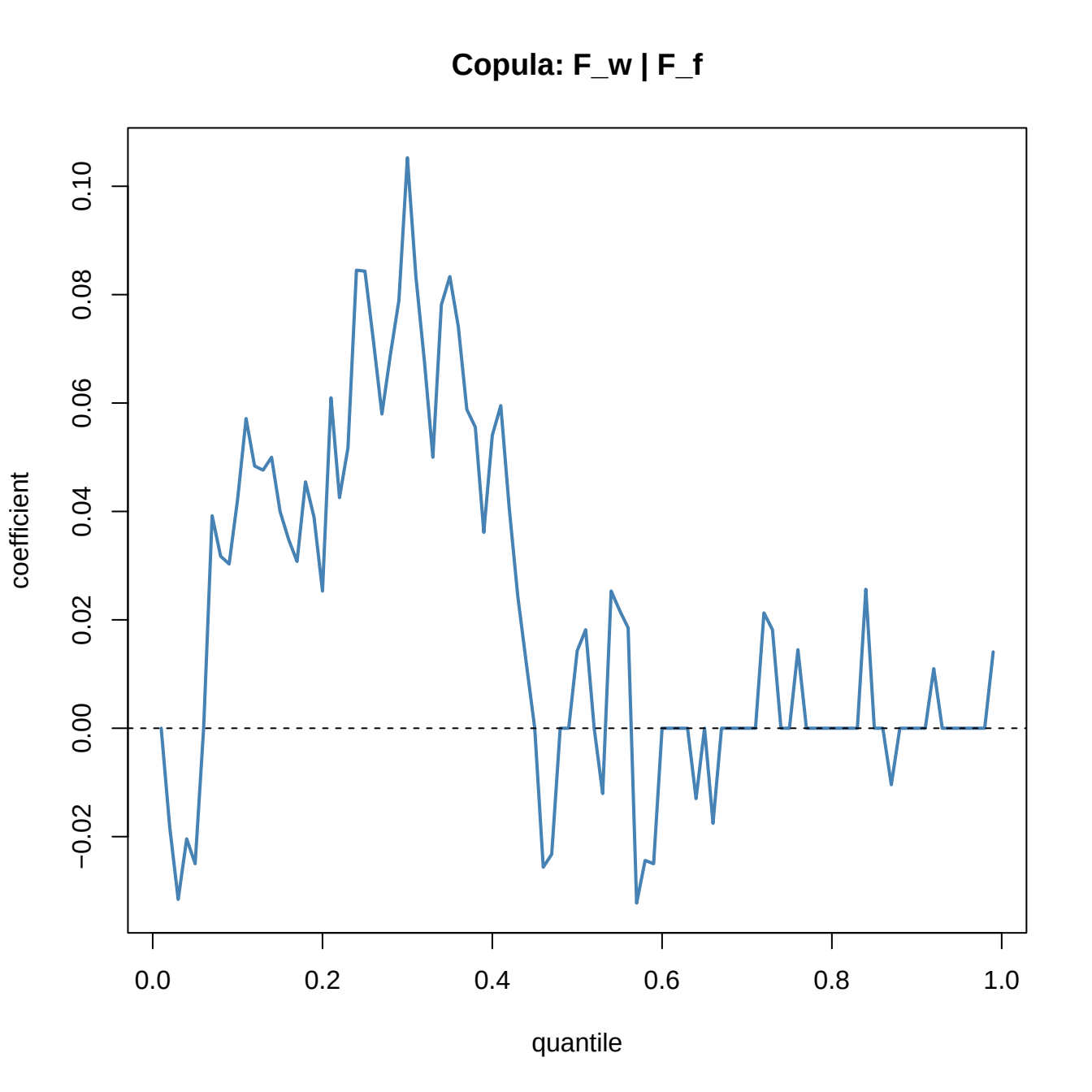

OLS−CUSUM of equation f_p
Empirical fluctuation process
1.0
0.0
−1.0
0.0
0.2
0.4
0.6
0.8
1.0
Time

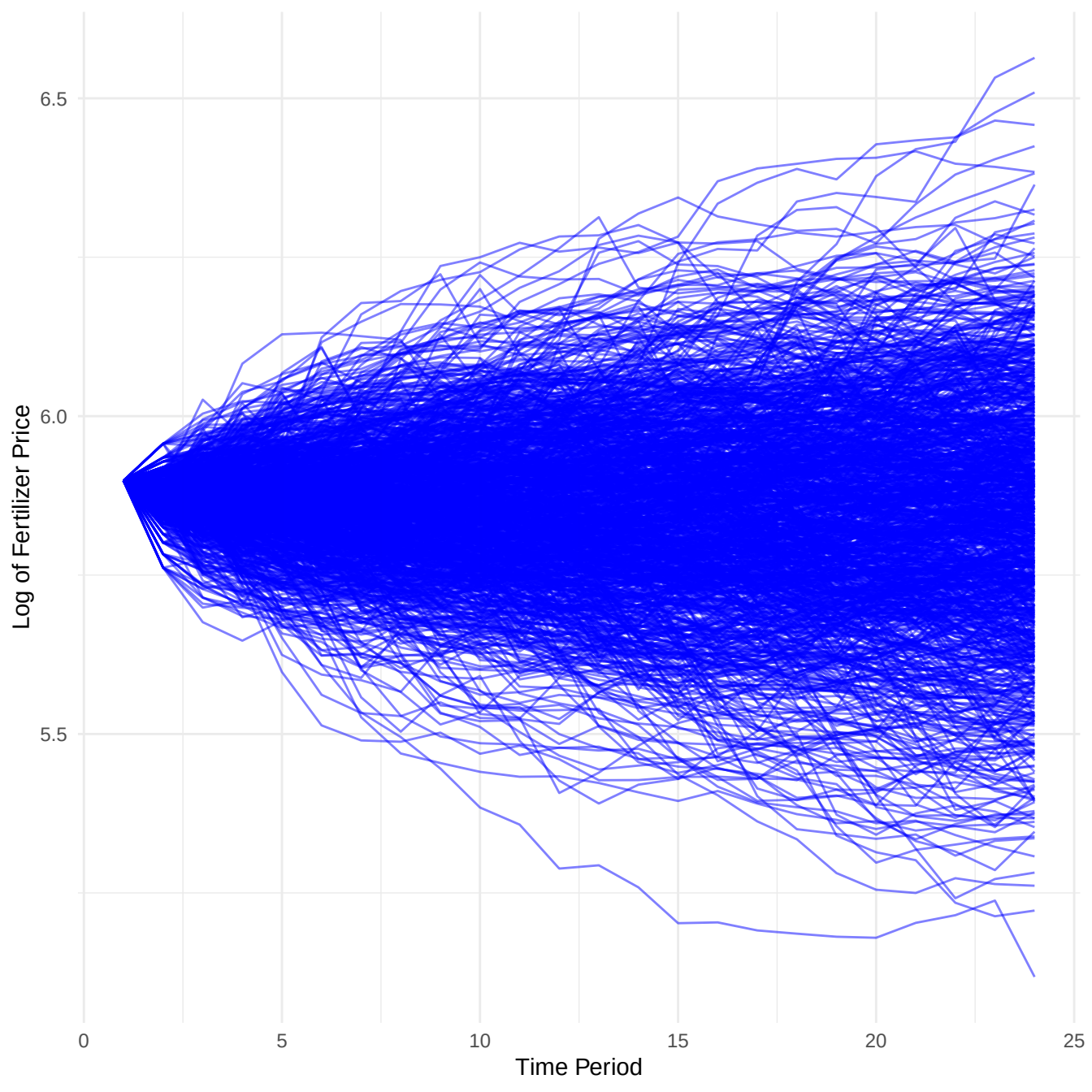

RE of equation w_p
Empirical fluctuation process
0.0
0.5
1.0
1.5
0.0
0.2
0.4
0.6
0.8
1.0
Time

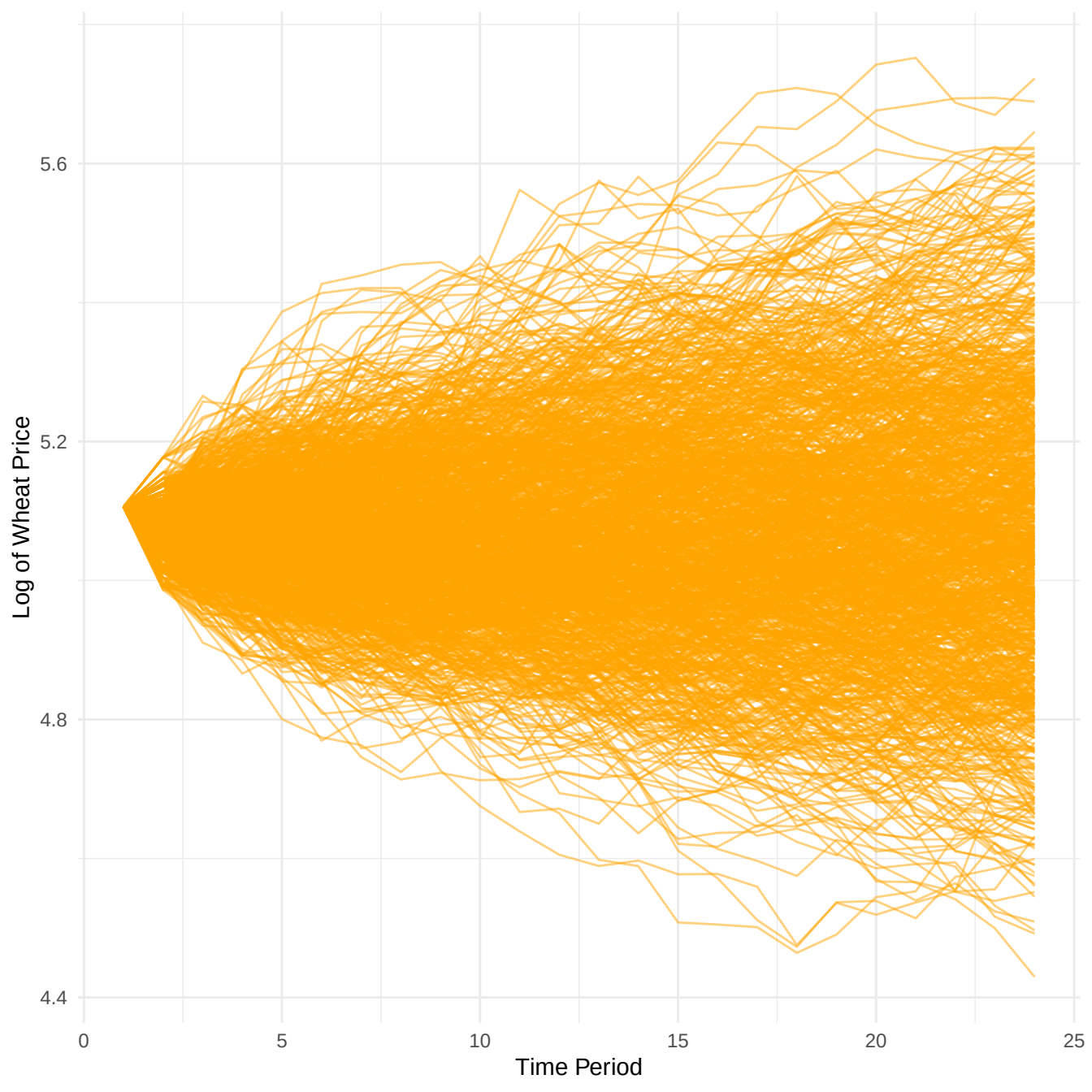

RE of equation f_p
Empirical fluctuation process
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Time

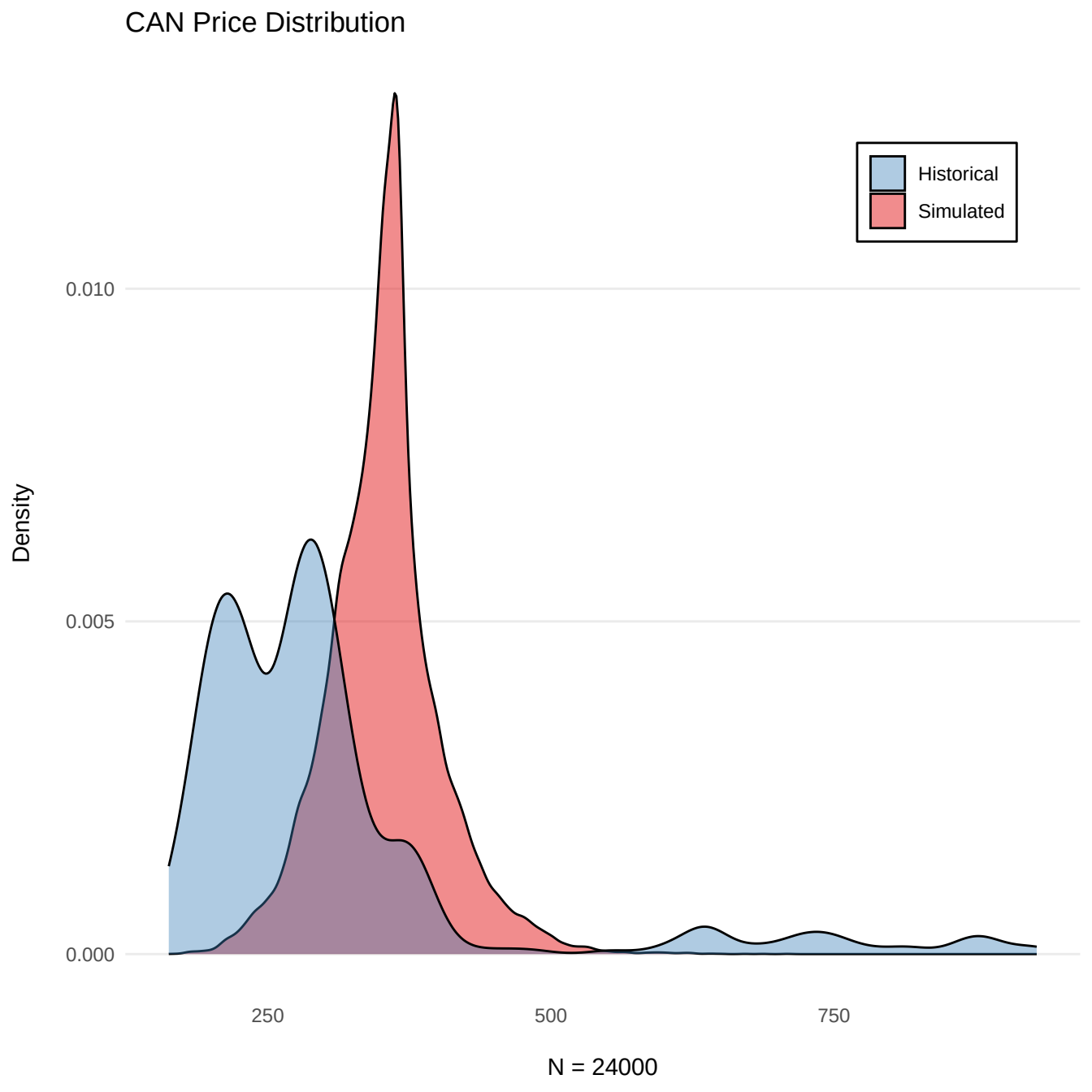

BIC and Residual Sum of Squares
BIC
RSS
−1200
−1210
−1220
−1230
−1240
−1250
−1260
0.92
0.90
0.88
0.86
0.84
0.82
0.80
0
1
2
3
4
5
Number of breakpoints

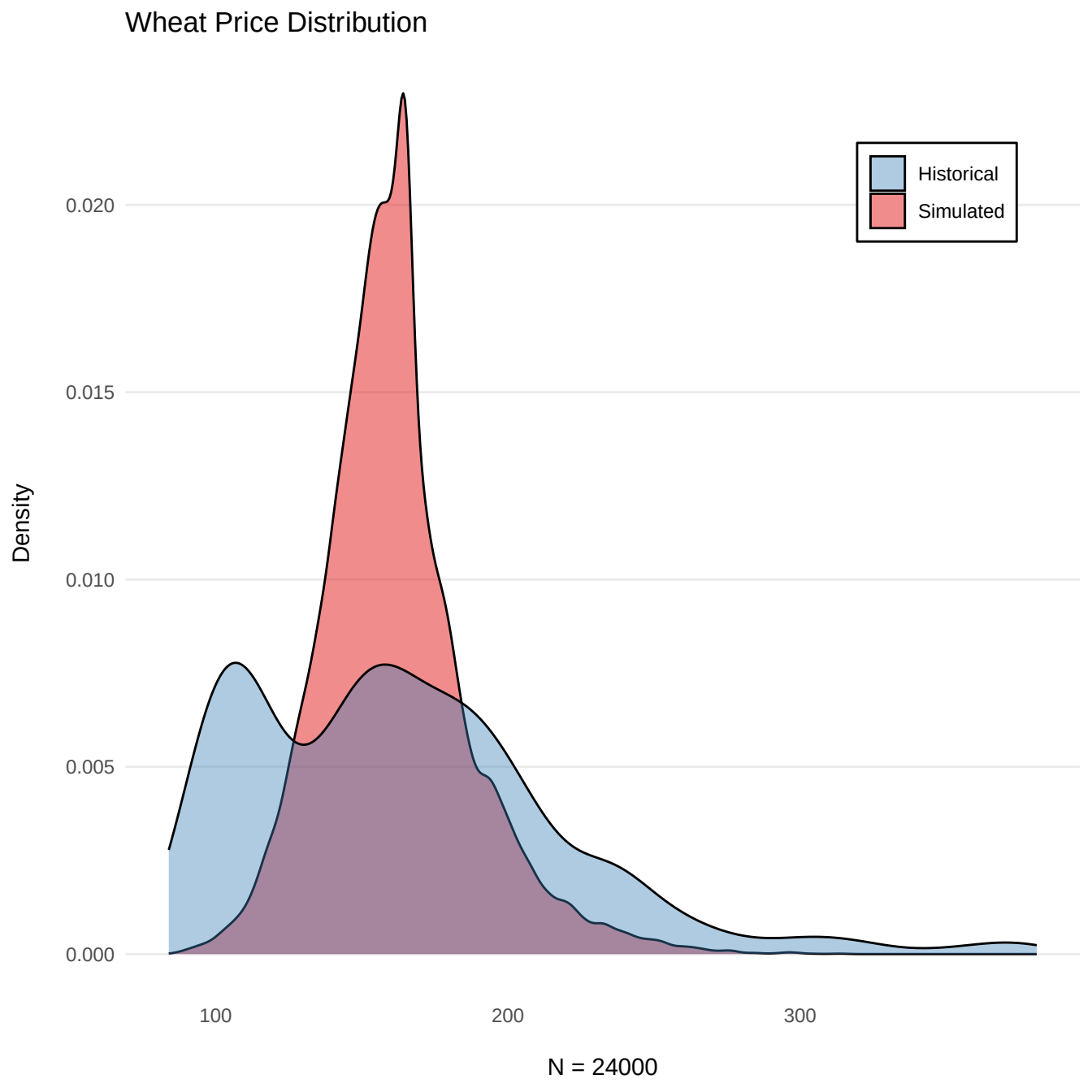

BIC and Residual Sum of Squares
BIC
RSS
−1280
−1300
−1320
−1340
−1360
0.70
0.69
0.68
0.67
0.66
0
1
2
3
4
5
Number of breakpoints

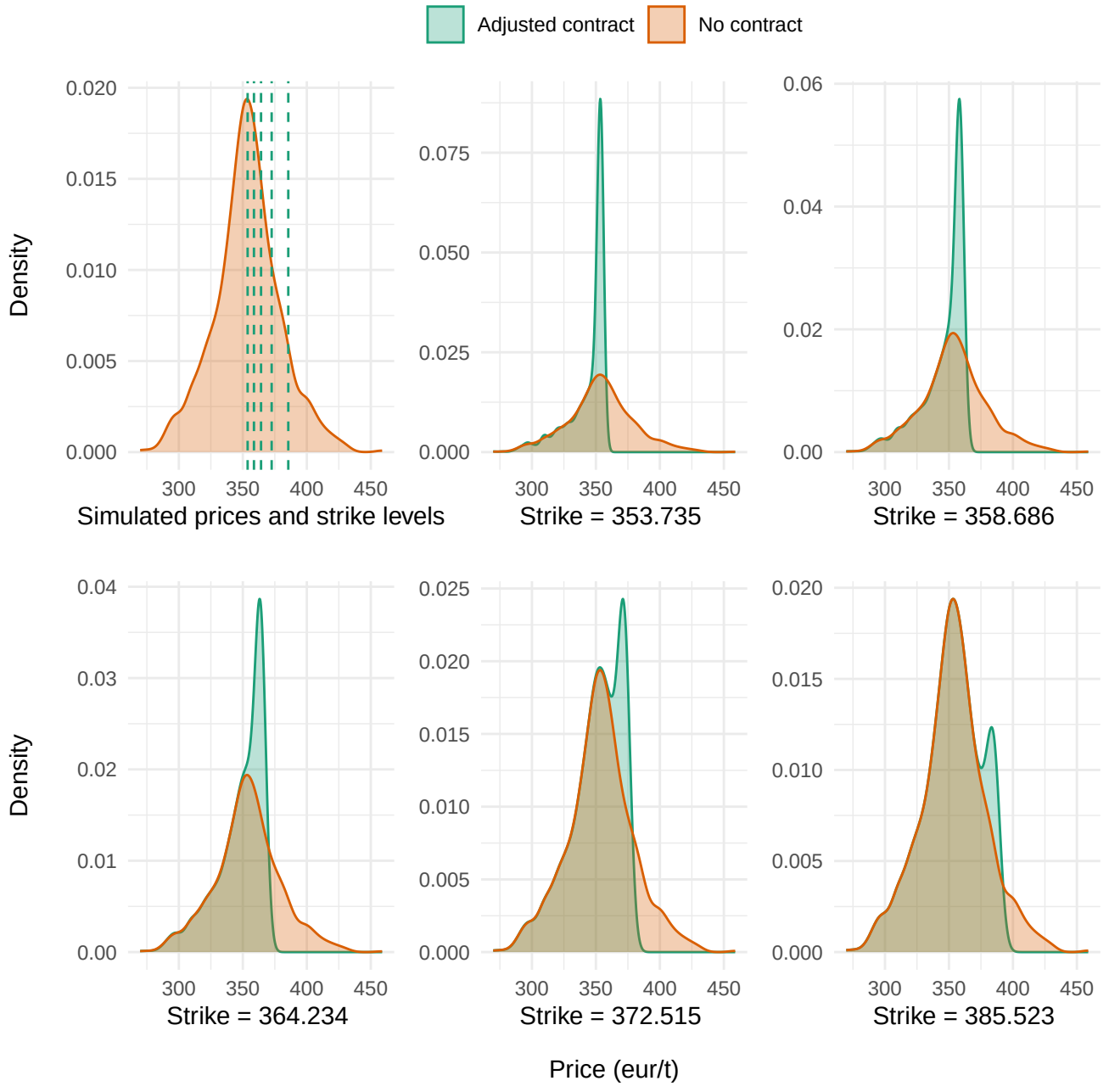

Copula: F_w | F_f
coefficient
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
0.0
0.2
0.4
0.6
0.8
1.0
quantile

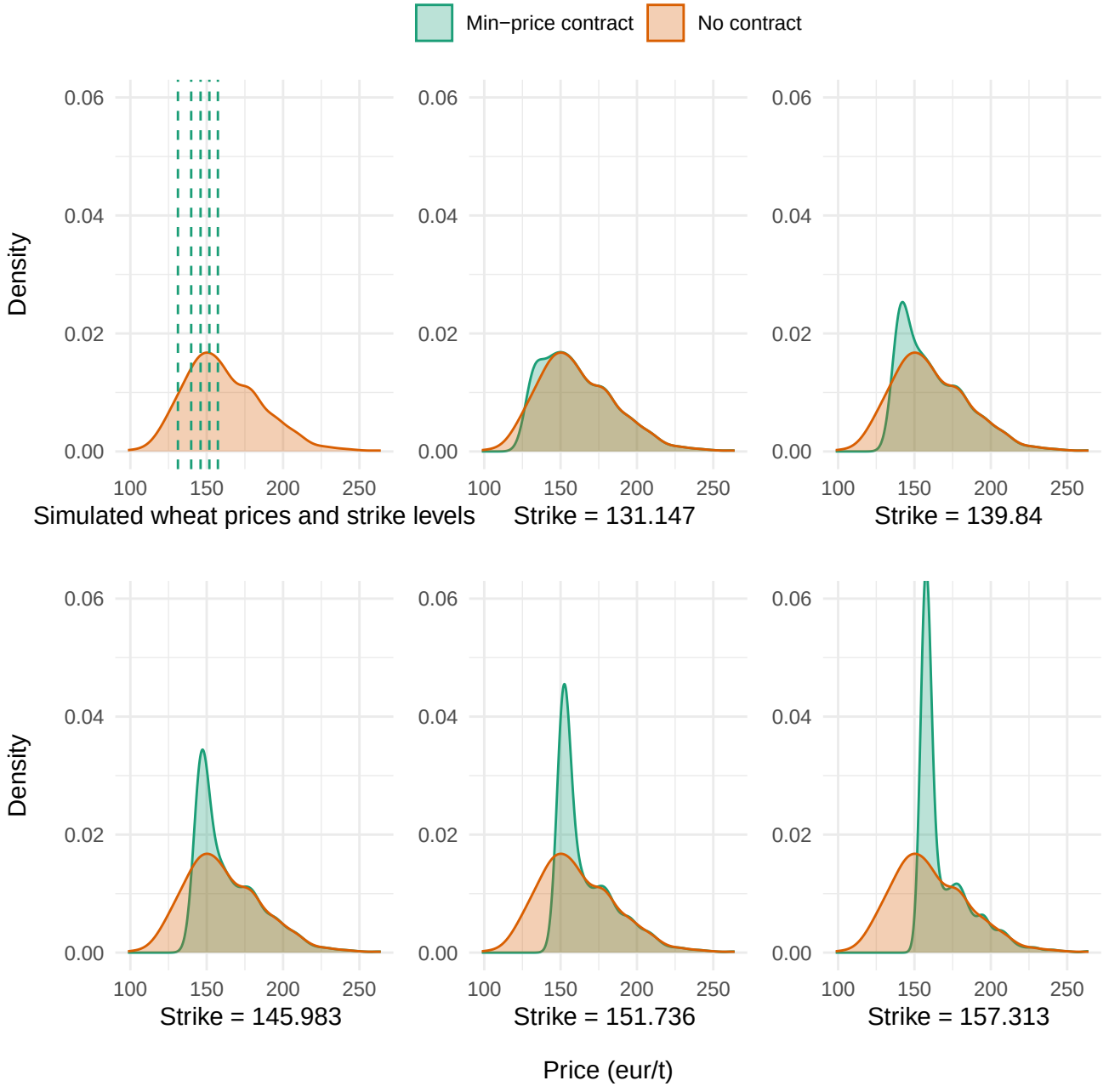

6.5
6.0
5.5
Log of Fertilizer Price
0
5
10
15
20
25
Time Period

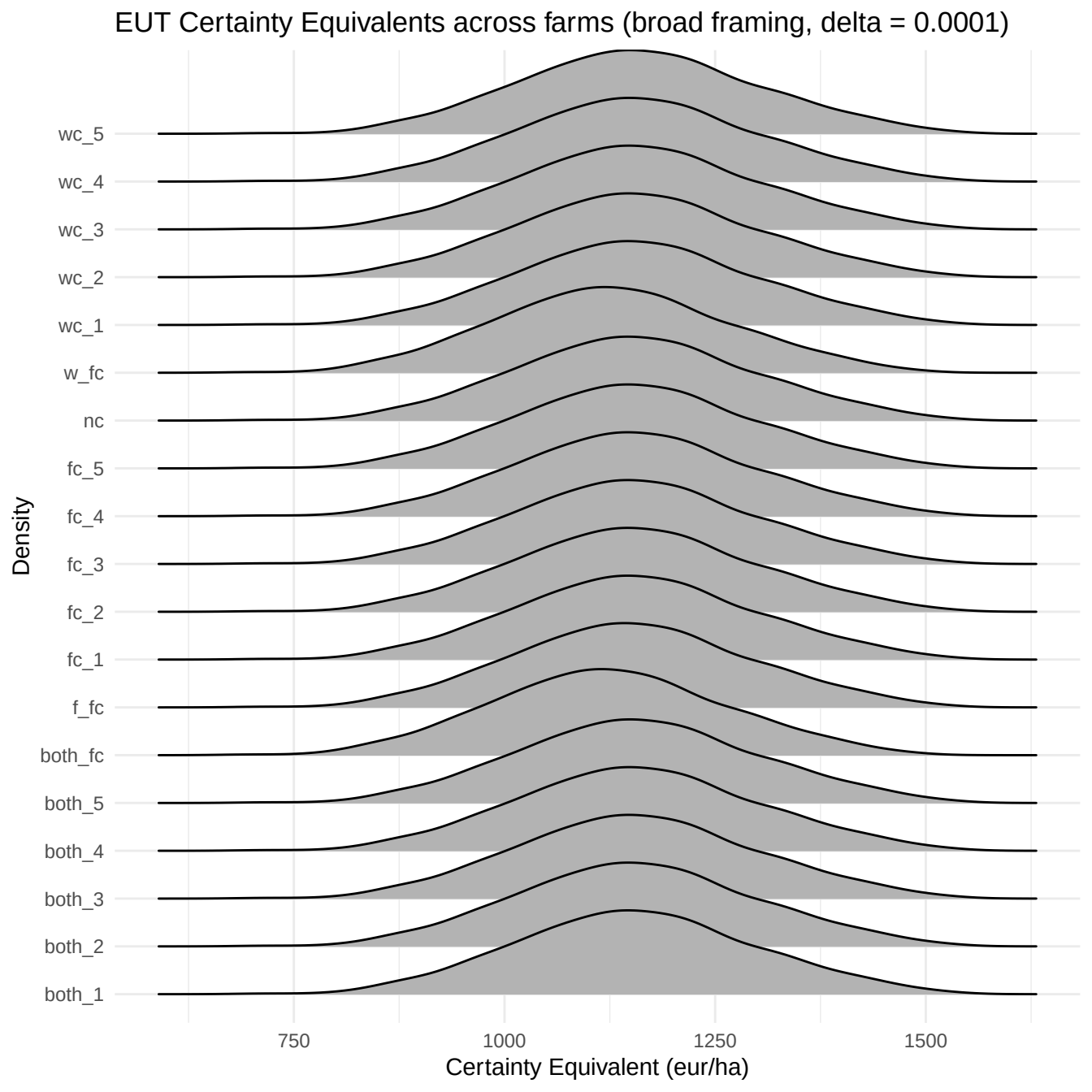

5.6
5.2
4.8
4.4
Log of Wheat Price
0
5
10
15
20
25
Time Period

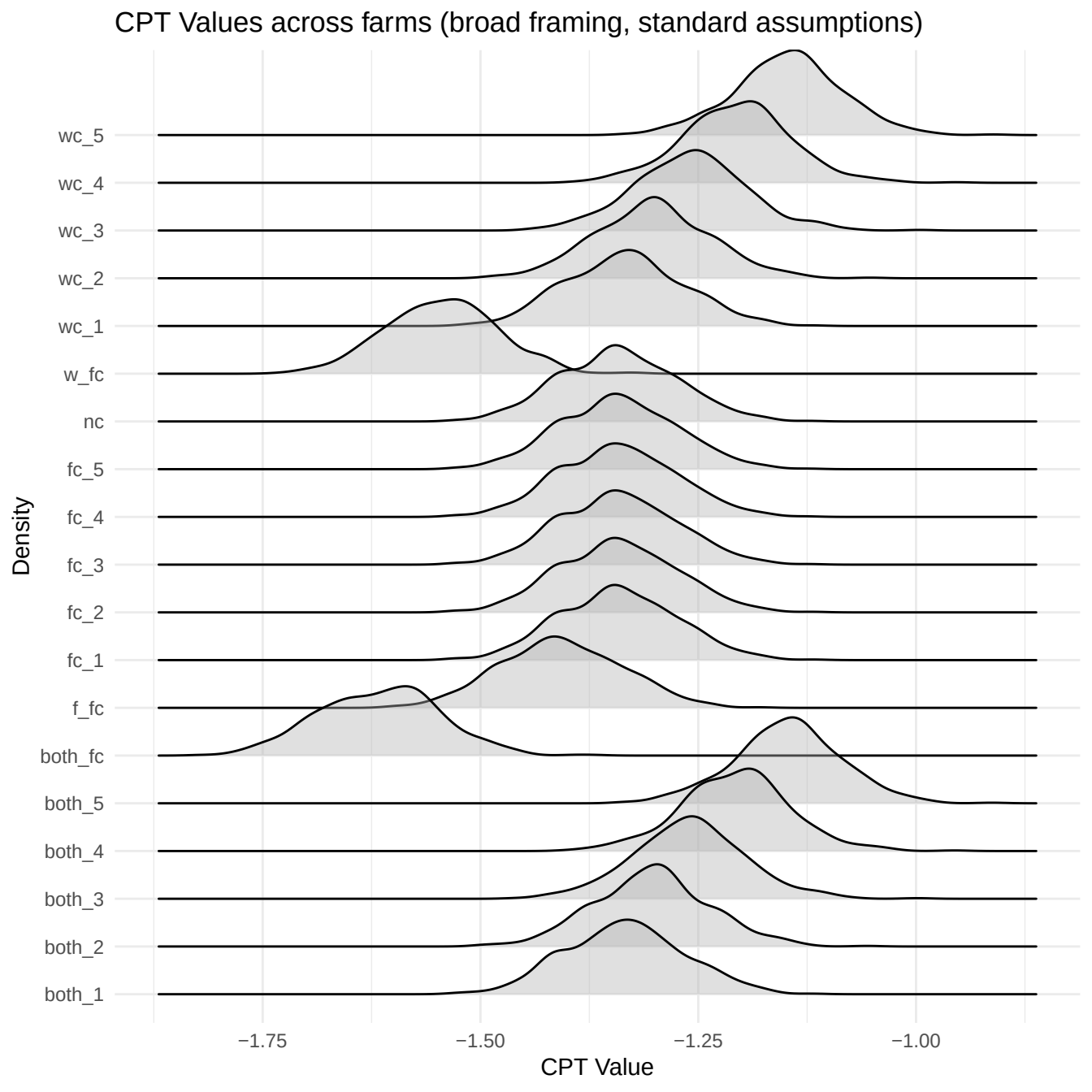

CAN Price Distribution
Historical
Simulated
0.010
0.005
0.000
Density
250
500
750
N = 24000

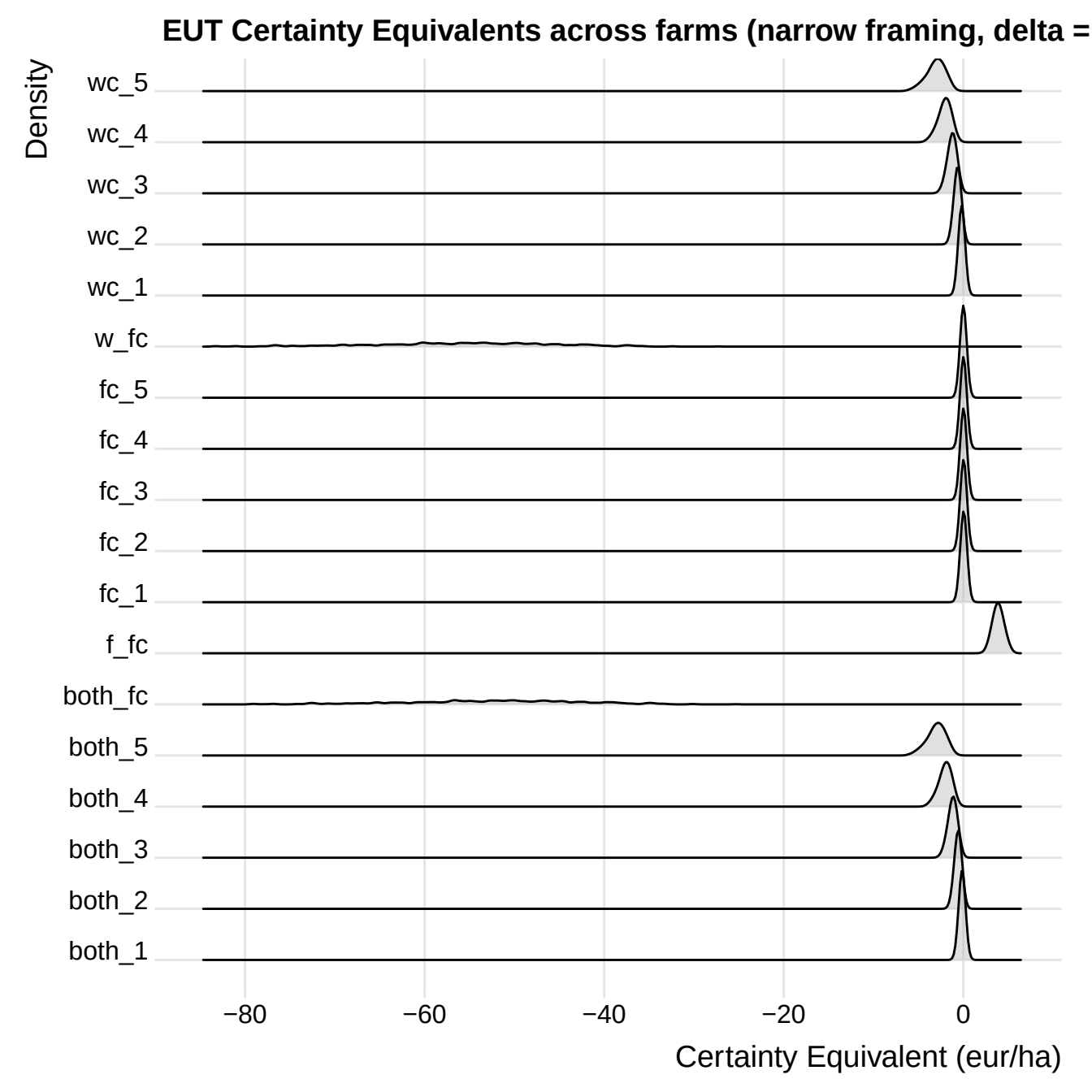

Wheat Price Distribution
Historical
Simulated
0.020
0.015
0.010
0.005
0.000
Density
100
200
300
N = 24000

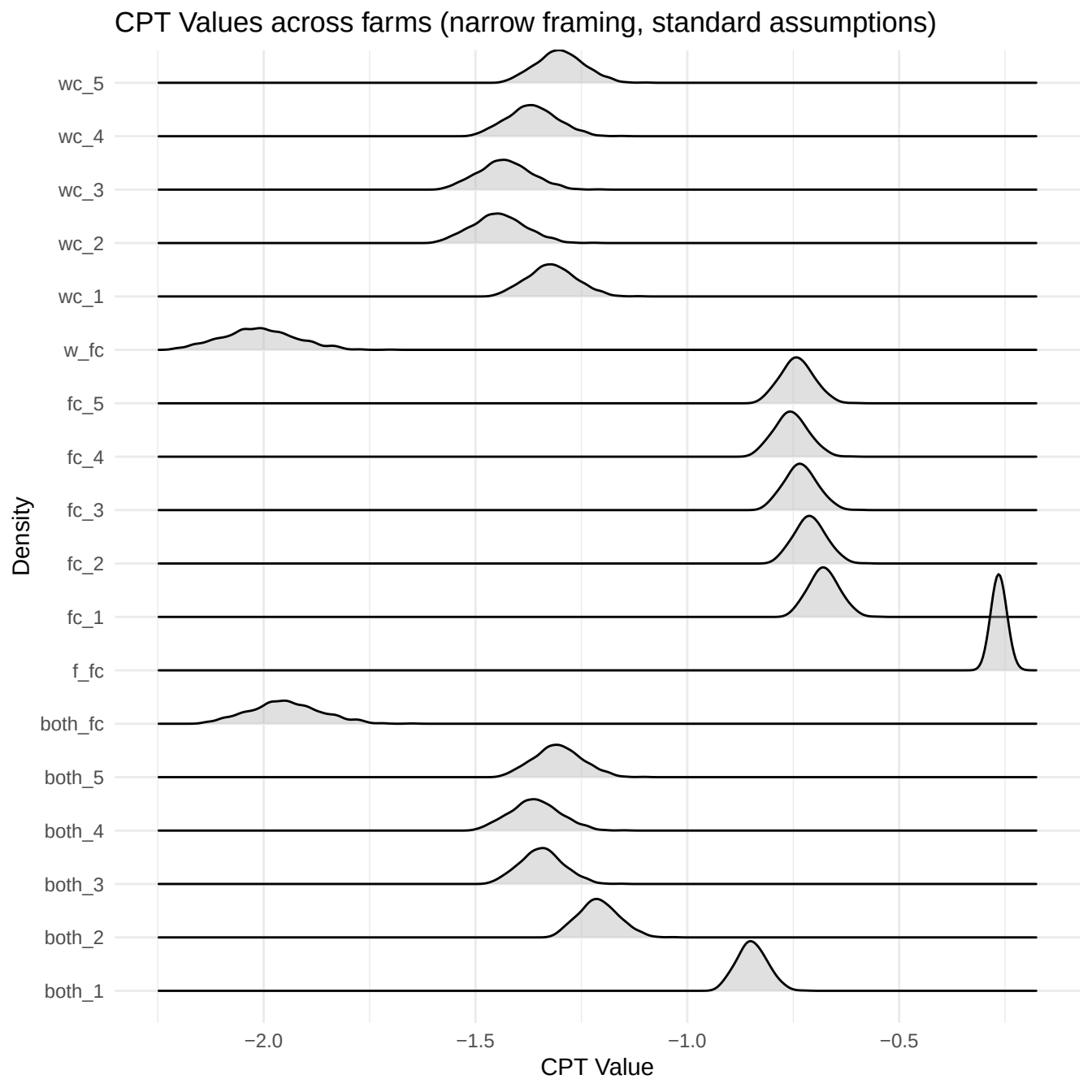

Adjusted contract
No contract
Density
Simulated prices and strike levels
Strike = 353.735
Strike = 358.686
Strike = 364.234
Strike = 372.515
Strike = 385.523
Price (eur/t)


Min−price contract
No contract
Density
0.06
0.04
0.02
0.00
100
150
200
250
Simulated wheat prices and strike levels
Strike = 131.147
Strike = 139.84
Strike = 145.983
Strike = 151.736
Strike = 157.313
Price (eur/t)


EUT Certainty Equivalents across farms (broad framing, delta = 0.0001)
wc_5
wc_4
wc_3
wc_2
wc_1
w_fc
nc
fc_5
fc_4
fc_3
fc_2
fc_1
f_fc
both_fc
both_5
both_4
both_3
both_2
both_1
Density
750
1000
1250
1500
Certainty Equivalent (eur/ha)


CPT Values across farms (broad framing, standard assumptions)
wc_5
wc_4
wc_3
wc_2
wc_1
w_fc
nc
fc_5
fc_4
fc_3
fc_2
fc_1
f_fc
both_fc
both_5
both_4
both_3
both_2
both_1
Density
−1.75
−1.50
−1.25
−1.00
CPT Value


EUT Certainty Equivalents across farms (narrow framing, delta =
Density
wc_5
wc_4
wc_3
wc_2
wc_1
w_fc
fc_5
fc_4
fc_3
fc_2
fc_1
f_fc
both_fc
both_5
both_4
both_3
both_2
both_1
−80
−60
−40
−20
0
Certainty Equivalent (eur/ha)


CPT Values across farms (narrow framing, standard assumptions)
wc_5
wc_4
wc_3
wc_2
wc_1
w_fc
fc_5
fc_4
fc_3
fc_2
fc_1
f_fc
both_fc
both_5
both_4
both_3
both_2
both_1
Density
−2.0
−1.5
−1.0
−0.5
CPT Value